

CAPITALAND RETAIL MANAGEMENT PTE LTD
Third Party POS System Integration Specifications
(POS Acceptance Form - Option A)

For Tenant who has accepted and opted for Option A of the POS Acceptance form for their own Point of Sale (POS) terminal or server to integrate with CapitaMall Trust (CMT)'s POS to transmit total daily sales in text format through file transfer protocol (FTP) transmission to CMT's server, the Third Party POS System Integration Specification shall be used by Tenant's POS Vendor to carry out the necessary development and implementation according to the specifications.

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A System Requirements

1. The designated Points-of-Sales (hereafter POS) terminal or server should be running on an operating system (Windows 98, 2000, XP etc) with support for TCP/IP network stack.
2. The Tenant must ensure that there is a dedicated network card (Ethernet adapter with RJ-45) in the POS terminal or server to facilitate successful connection to the Mall network using Local Area Network (LAN) point.
3. One LAN point using UTP cabling will be laid into Tenant outlet by our Cabling Contractor. The exact location of cable termination point to be coordinated with the tenant. During the cabling works, Tenant representative has to be onsite to supervise and confirm how this data point should be laid to the Tenant's POS terminal or server. Any additional cost subsequently incurred by Tenant or Tenant's POS vendor arising from this cabling work will be borne by the tenant.
4. Tenant's POS vendor should develop a function or program in the POS terminal or server to **automatically** generate and transmit the Daily Transaction File to the CMT's server after the end of each business day (One transaction for each business day). Provision must be made to allow for backlog processing of delayed or failed transmission (e.g. manual trigger). The method of file transfer is FTP. Refer to Section C and D for Transaction File Naming Convention Specification and Daily Transaction File Format Specification respectively.
5. If there are more than 1 POS terminals in the Tenant's outlet, the sales transactions of all POS terminals must be consolidated into 1 file at one designated PC / Server / POS terminal before the file is transmitted to the CMT's server.
6. Tenant should appoint a dedicated staff or their POS Vendor to coordinate with DCS Synthesis' Project Co-coordinator for connecting of their POS terminal or server into the Mall network and the testing of the system integration.

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B Daily Interface Operation

1. Tenant shall generate 1 Daily Transaction File in a text file format after every end of day closing and **automatically** transmit the Daily Transaction File to the CMT's server. The IP address will be provided by the Mall together with required user ID and password to control the access to the Mall server; including Machine ID and the Daily Transaction Filename.
2. Tenant can choose to transmit their Daily Transaction File between 9pm to the next day 11.59am.
3. If the Tenant POS terminal or server is not able to send any day's Daily Transaction File successfully due to whatever reason, the Tenant has to send any backlog daily transaction files (e.g. manual trigger of "export and send" process from the POS system) to the CMT's server upon resolving the problem.
4. If the Tenant's POS terminal or server is not operational due to the POS technical problem and is not able to FTP data from the outlet, the Tenant representative has to update the CMT Management immediately.

If you have any query, please contact the following:

DCS Synthesis Pte Ltd	Mr Jimmy Teo	9069 2944 (HP) 6282 7655 (O)
CapitaLand IT	Mr Randell Low	6826 5720 (O)

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C Transaction File Naming Convention Specifications

1. The file naming convention follows the DOS filename standard (8.3 format) for backward compatibility to DOS-based POS terminals.

Position	Size	Description
1	1 char	Document Type “D” for Daily Transaction File
2 to 8	7 char	Machine ID This is a unique identifier assigned to individual POS System provided by the mall. Format of Machine ID is made up of 7 numeric characters. Example of Machine ID: 1321000
9	1 char	Constant “.”
10 to 12	3 num	File Serial Number 3 numerals at the end of the filename to differentiate each daily file sent to the FTP server. Example: <filename>.XXX “XXX” shall be a running number incremented by 1 each time a file is generated and sent. The number should start from “000” and once it reaches “999”, the number will be reset to “000” and increase from there again.

2. Machine ID should not be hard coded in the POS software system or Cash Register. It should be a configurable parameter in a database table or in an “ini” file
3. Daily Transaction filename must start with a “D” for each exported file. Example of a filename:
D1321000.135

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D Daily Transaction File Format Specification

1. The exported file shall be a text file with fixed-length fields in the following format:

Field No.	Type Size	Position	Field	Description
1	Character (1)	1	Document Type	"D" for Daily Transaction File
2	Numeric (7)	2 - 8	Machine ID	Unique identifier assigned to individual POS System across the mall. Format of Machine ID is made up of 7 numeric characters and this ID is assigned to the tenant by the Mall Example of Machine ID: 1321000
3	Date (8)	9 - 16	Transaction Date	Date of sales transaction adopting "YYYYMMDD" format. Example: 20050928 (28 Sept 2005)
4	Numeric (11) 99999999.99	17 - 27	Sales Amount	This is the defined as: Gross Sales less Discount plus Service Charges (if any) before tax (i.e. GST or CESS)

2. Machine ID should not be hard coded in the POS software system or Cash Register. It should be a configurable parameter in a database table or in an "ini" file

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E Sample Transactions

Example A

For Retail tenant (Pay by Cash, Nets or Credit card):

A Shopper purchase S\$200.00 of goods and get S\$ 10% discount.
Final he paid S\$ 180.00 + 5% GST

Receipt Example:

Sales item	S\$200.00
Discount (10%)	S\$ 20.00
GST	S\$ 9.00
Total	S\$189.00

Field	Data
Machine ID	1321000
Transaction Date	20050928 (means 28 th Sept 2005)
Sales Amount	180.00

Sample Filename: D1321000.000

Sample Data: D132100020050928000000180.00

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Example B

For F&B Restaurant (Pay by Cash, Nets or Credit card):

A Shopper eats in a Restaurant and spends S\$200.00 and get S\$ 10% discount.
Final he paid S\$ 180.00 + 5% GST + 10% Service Charge + 1 % CESS

Receipt Example:

Sales item	S\$200.00
Discount (10%)	S\$ 20.00
Service charge	S\$ 18.00
CESS	S\$ 1.80
GST	S\$ 9.00
Total	S\$208.80

Field	Data
Machine ID	1321000
Transaction Date	20050928 (means 28 th Sept 2005)
Sales Amount	198.00

Sample filename: D1321000.000

Sample Data: D13210002005092800000198.00